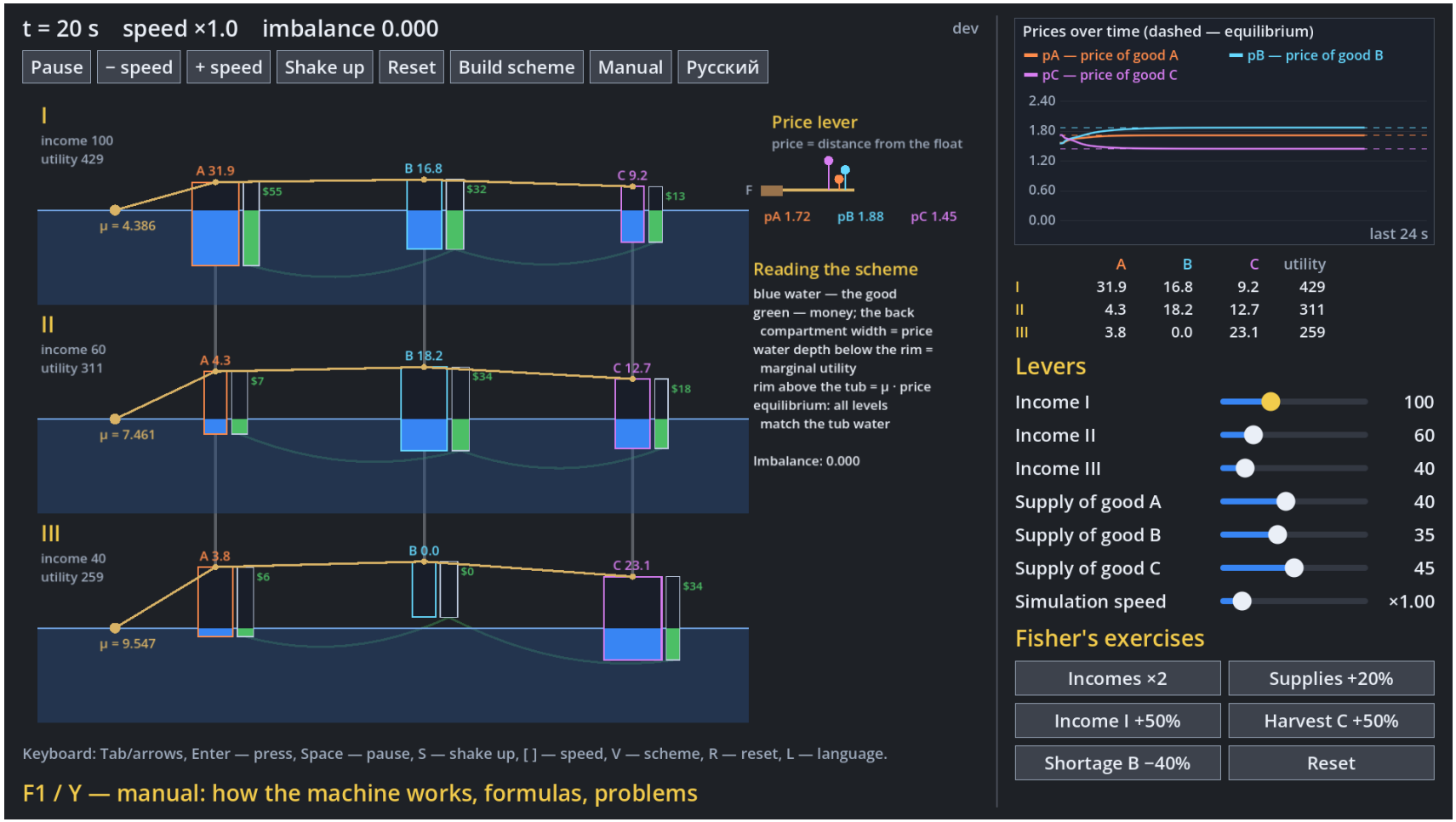
The Fisher Machine

A hydrostatic machine for computing equilibrium prices (1891): how it works, formulas, problems



The program screen. F1 or Y on the gamepad opens/closes this manual, ← / → or LB / RB flip pages.

1. What it is

In 1891 Irving Fisher defended his Yale thesis “Mathematical Investigations in the Theory of Value and Prices” (published in 1892). In it he built — independently of Walras — a general equilibrium model: how prices form in all markets at once when each buyer has an income and tastes of his own. He also described a machine of cisterns, levers and floats that finds these prices — with water.

Fisher, a student of the physicist Gibbs, built the machine and, by his account, showed it to students for about 25 years; in 1925 he built a second one. Both are lost. In 2005 the economists William Brainard and Herbert Scarf reconstructed from Fisher's description how the mechanism works and simulated it on a computer — this program follows their reconstruction.

A companion to MONIAC: the Phillips machine (1949) pumps flows — income, taxes, spending of a whole economy. The Fisher machine answers another question: what the prices of goods will be and who buys how much. The first is Keynesian macroeconomics, the second microeconomics and general equilibrium.

2. A cistern is a buyer's taste

Fisher's key invention is a cistern of a special shape. Pour x water into it (x is how much of the good the buyer got): the depth of the surface below the rim equals the **marginal utility** u′(x) — how much one more unit is worth to him. An empty cistern has the greatest depth, the good is desired; a full one has zero depth, the buyer is satiated.

In the program the cisterns are rectangular (as, judging by photographs, in the second machine of 1925): width w and depth d give u′(x) = d − x/w and utility U = d·x − x²/(2w). The wider the cistern, the slower the good gets boring; the deeper, the more it is desired.

Consumer	A	B	C	Income
I	4 × 12	3 × 10	2 × 8	100
II	2 × 9	4 × 12	3 × 10	60
III	3 × 10	2 × 8	5 × 12	40
Supply	40	35	45	

Default cisterns “width × depth”. I likes A, II likes B, III likes C; I has the largest income.

3. How the machine works

Nine cisterns — one per “consumer × good” pair — float in a common tub. Each is divided by a partition into two compartments.

1. **Front compartment** (thickness 1) — the quantity of the good. The front compartments of one good for all consumers are joined by a tube: they hold exactly the whole supply of that good, $\Sigma_i x_{ij} = E_j$. Water flows to whoever has the lower level.
2. **Back compartment** — money. Its thickness equals the price p_j (walls of flexible leather), so it holds $x \cdot p$ water — the cost of the purchase. The back compartments of one consumer are joined by a tube: they hold his whole income, $\Sigma_j y_{ij} = Y_i$.
3. **Consumer levers** hold the rims of his cisterns at heights proportional to prices: $h_{ij} = \mu_i \cdot p_j$. This is the condition “marginal utilities are proportional to prices”.
4. **The price lever** lies on the water: float F and pins A, B, C. The thickness of the back compartments is proportional to the pin's distance from the float — this is the price.

If a cistern has more water than it “should”, it is heavier than the water it displaces and sinks, the lever goes down; if less, it rises. Water flows through the tubes slower than the levers move. The machine comes to rest only when the level in **every** compartment matches the tub water — that is equilibrium. “Imbalance” on screen is the largest mismatch of levels.

As in Brainard and Scarf, the mechanics are simplified: ideal massless levers moving in proportion to the force; water flows in proportion to the level difference. The equilibrium itself is exactly Fisher's.

4. Fisher's equations

The machine solves the system we now call exchange equilibrium:

$$u'_{ij}(x_{ij}) = \lambda_i \cdot p_j \;\; \Sigma_j \; p_j \cdot x_{ij} = Y_i \;\; \Sigma_i \; x_{ij} = E_j$$

The first: everyone spends so that the last dollar brings the same benefit on every good (λ_i is the marginal utility of money; if a good is too expensive, x = 0). The second: everyone spends exactly his income. The third: what is bought equals the supply.

With the defaults: $p_A = 1.722$, $p_B = 1.879$, $p_C = 1.453$. The program solves these equations itself and draws the answer dashed on the price chart — the machine should arrive there.

Consumer	x_A	x_B	x_C	λ	Utility
I	31.9	16.8	9.2	2.334	429
II	4.3	18.2	12.7	3.971	311
III	3.8	0.0	23.1	5.082	259

III buys no B at all: at this price his last unit of income is better spent on C.

5. Screen and controls

Left: three bands — consumers I, II, III, each with cisterns for goods A, B, C, and to the right of each its back compartment (green water, width = price). Above the tub water is the cistern rim, the yellow line is the consumer's lever. Further right is the price lever. The right panel: the price chart with the dashed equilibrium, a table of purchases and utilities, levers (incomes and supplies) and Fisher's exercises.

Gamepad (Steam Deck)	Keyboard	Action
D-pad / left stick	arrows, Tab	move between buttons and sliders
← / → on a slider	← / →	change an income or supply (1/40 of the scale)
A	Enter	press a button
X	Space	pause / run
Start	S	shake up: water split evenly, the machine searches again
Y	F1	manual
Select	L	language
LB / RB	[/]	speed ÷2 / ×2
—	R	reset

Incomes and supplies change as with Fisher's plungers: water is added to the compartments proportionally and the machine moves to the new equilibrium before your eyes.

6. Fisher's exercises

In the thesis Fisher discusses eight “experiments” on the machine. The panel buttons reproduce five of them; each starts from the defaults, but the water is not shaken up — you see the machine move over.

Exercise	p_A	p_B	p_C	U_I	U_{II}	U_{III}
Baseline	1.722	1.879	1.453	429	311	259
Incomes ×2	3.444	3.758	2.905	429	311	259
Supplies +20%	1.458	1.591	1.170	467	356	303
Income I +50%	2.203	2.393	1.736	466	261	224
Harvest C +50%	1.748	1.920	0.938	438	340	336
Shortage B −40%	1.814	2.758	1.545	392	260	247

Incomes ×2. All prices exactly double (3.444 = 2 · 1.722), while purchases and utilities stay the same: money by itself does not make anyone richer.

Supplies +20%. Mill believed prices would fall in the same proportion, by a factor of 1.2. The machine shows otherwise: A changes by −15 %, C by −19 %.

Income I +50%. The rich I buys more, prices rise — II and III, whose income did not change, lose (−16 % and −14 % utility).

Harvest C +50% and **shortage B −40%** — in problems 2 and 3.

7. Problems: from life to cisterns

For each problem: 1) decide what changed — incomes (the “Income” levers) or supplies (the “Supply” levers); 2) set them and watch the machine move to the new dashed line; 3) compare prices and utilities in the table on the right. Units are notional: directions and proportions matter.

Problem 1. The printing press

Problem. The state printed money and gave everyone the same percentage — incomes doubled. Did people get richer? Fisher later became famous for the quantity theory of money MV = PT — the machine shows its essence.

Quantity	Calculation	On the machine
Change	$Y_i \times 2$	“Incomes ×2” button
Prices	p × 2 = (3.44; 3.76; 2.91)	the dashed lines double
Purchases	the same	table on the right

Solution. All prices double, purchases and utilities do not change. The back compartments became twice as thick (price ×2) and hold twice the money — while the front ones hold the same goods.

Conclusion: if everyone gets the same extra money, only prices grow. People get richer only if there are more goods — or if the money is not shared equally (problem 4).

Problem 2. A bumper harvest

Problem. The harvest of good C (say, potatoes) is one and a half times the usual. How much cheaper does it get? Who wins — only potato lovers?

Quantity	Calculation	On the machine
Change	$E_C = 45 \to 67$	“Harvest C +50%” button
Price of C	$1.453 \to 0.938$	−35 %
Prices of A and B	1.748; 1.920	almost unchanged

Solution. C becomes 35 % cheaper — more than a third, though there is half as much again. Everyone's C cisterns fill up and the marginal utility of potatoes falls.

Everyone wins: utility of I +2 %, II +9 %, III +30 %. III, who likes C, wins most, but the others too: less money goes on potatoes and more is left for A and B.

Problem 3. A crop failure

Problem. The supply of good B (say, grain) fell by 40%. How much does its price rise? Who suffers more — the one who likes it, or the poorer one?

Quantity	Calculation	On the machine
Change	$E_B = 35 \to 21$	“Shortage B −40%” button
Price of B	1.879 → 2.758	+47 %
Utilities	392; 260; 247	was 429; 311; 259

Solution. The price of B rises by 47 %. II, who likes B, loses most: −17 %. I loses −9 % — he is rich and outbids. III, who did not buy B anyway, loses least: −5 % — through the dearer A and C.

Conclusion: a shortage hits hardest those for whom the good matters; the general price rise touches everyone.

Problem 4. A tax on the rich

Problem. A tax of 20 is taken from I and given to the poorest, III. Who wins and who loses? Can someone the tax does not touch at all win?

Quantity	Calculation	On the machine
Change	Y = (100; 60; 40) → (80; 60; 60)	“Income I” and “Income III” sliders
Prices	(1.705; 1.756; 1.563)	A and B get cheaper, C dearer
Utility of II	311.0 → 315.3	+1 %

Solution. III wins a lot (+31 %), I loses (−12 %). But II, whose income did not change, also wins: +1 %. Money went from an A lover to a C lover — and B, which II needs, got cheaper than anything.

This is one of Fisher's exercises: a change in someone else's income can make a third person better off — through prices.

Problem 5. When does a good become free?

Problem. There is more and more of good A (a new technology). How does its price fall? At what supply does it become zero, like air? And what happens to the prices of B and C?

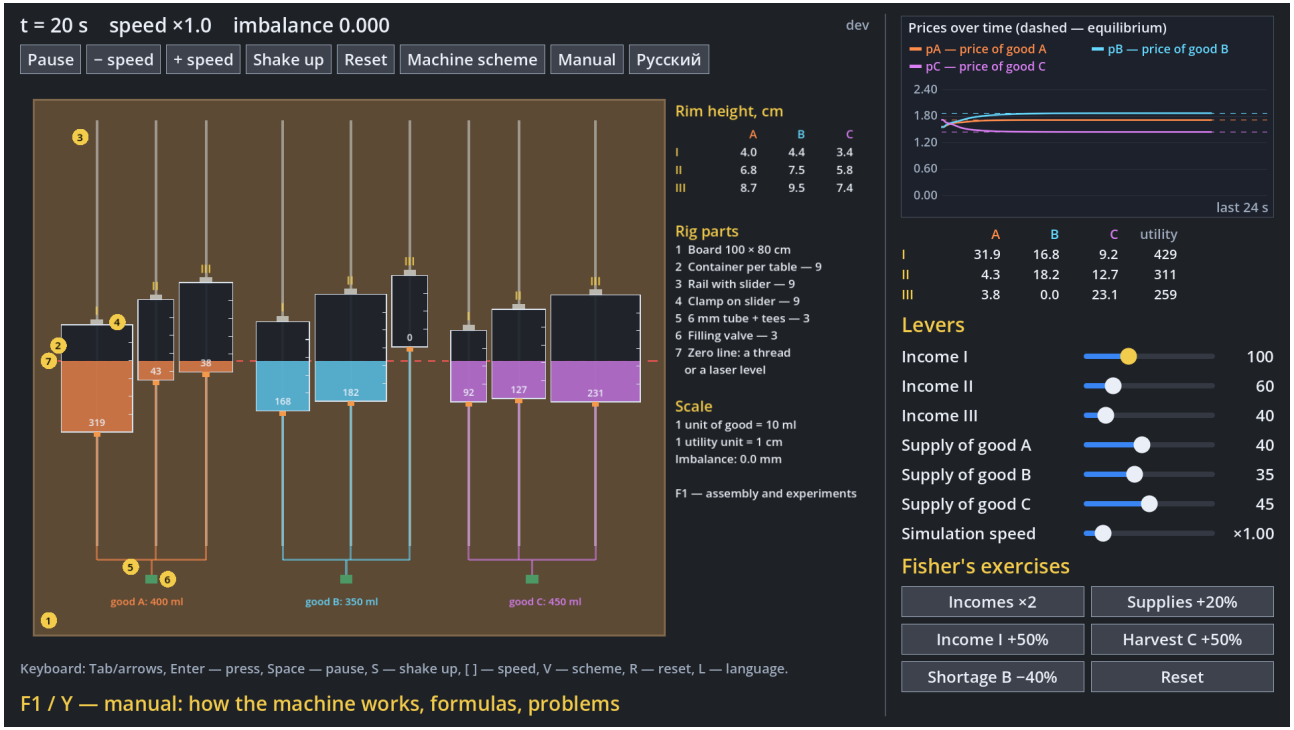
Quantity	Calculation	On the machine
Change	$E_A = 40 \to 60 \to 85$	“Supply of good A” slider
Price of A	$1.722 \to 1.118 \to 0.424$	
Price of B	$1.879 \to 2.026 \to 2.608$	gets dearer!
Satiation in A	$\Sigma\, w \cdot d = 96$	all A cisterns full

Solution. The price of A falls faster than supply grows: at 85 it is already 0.424. When supply reaches 96 — as much as all A cisterns hold together — people are satiated, nobody needs another unit, and the price becomes zero: the good is free, like air.

Meanwhile B and C get dearer ($1.879 \to 2.608$ for B): the money that used to go on A is now spent on them, and there is no more B and C than before.

8. Building the machine at home

The whole Fisher machine — floating cisterns, levers and leather bellows — cannot be repeated at home. But its heart — communicating vessels of a special shape — can be built in a weekend. The “Build scheme” button (V, L3 on the gamepad) shows the rig: nine containers on rails with sliders, the containers of one good joined at the bottom by a tube, a zero line playing the role of the tub water. The program takes over the levers and the money compartments: it tells you how high to set the rim of each container.



Why it works

Water in the containers of one good levels itself — the law of communicating vessels. If a rim is h cm above the line and the water stands on the line, the depth of water below the rim is h. By the container's design this depth is the marginal utility: so the water spreads so that everyone has $u'(x) = h = \lambda \cdot p$ — exactly Fisher's condition. If a container's bottom is above the water level, it stays empty: the buyer does not take that good (like III with good B).

Model	Rig
1 unit of good	10 ml
1 unit of marginal utility	1 cm of depth
Cistern w × d	container 2w × 5 × d cm
Tub water	zero line (thread, laser level)
Levers and bellows	rim heights from the program

Parts

#	Part	Qty	Notes
1	Board: plywood 100 × 80 cm on a stand	1	upright, levelled
2	Rectangular container	9	3 mm acrylic glued, or food boxes; sizes in the table below
3	Vertical rail with slider and scale	9	drawer slide or a lath + ruler
4	Container clamp on the slider	9	a wing screw to hold the height
5	Silicone tube 6 mm, tees, bottom fittings	3	one system per good, tubes below all bottoms
6	Filling valve (tee with a tap)	3	to fill and drain the good's water
7	Zero line	1	a stretched thread or a laser level

Also: three food colourings (one colour per good), a measuring cup, a 60 ml syringe for exact top-ups, a spirit level, a tray under the board.

Container sizes and the setting for the default equilibrium

Container	Inside: width × thickness × depth, cm	Holds, ml	Rim above line, cm	Water, ml
I · A	8 × 5 × 12	480	4.0	319
I · B	6 × 5 × 10	300	4.4	168
I · C	4 × 5 × 8	160	3.4	92
II · A	4 × 5 × 9	180	6.8	43
II · B	8 × 5 × 12	480	7.5	182
II · C	6 × 5 × 10	300	5.8	127
III · A	6 × 5 × 10	300	8.7	38
III · B	4 × 5 × 8	160	9.5	0
III · C	10 × 5 × 12	600	7.4	231

Width = 2·w, thickness 5 cm: then 1 cm of depth holds w units of the good, as in the model's cistern. Rims — from the program (the table on the right of the “Build scheme” screen).

Assembly and first experiment

1. Fix the 9 rails on the board in three groups — goods A, B, C, each with containers I, II, III. Level the board and stretch the zero line across all rails.

2. Glue fittings into the container bottoms and join each group with tubes through a tee with a tap. Run the tubes below all the bottoms — otherwise bubbles stay in them.

3. Set the container rims to the heights from the table above (or from the program screen after Reset).

4. Pour coloured water through the taps: A 400 ml, B 350 ml, C 450 ml (supplies × 10 ml) plus ≈ 34 ml for each system's tubes.

5. The water spreads by itself: all surfaces come to the zero line, and the volumes match the table. Container III · B stays empty — its bottom is above the line.

Experiments on the rig

Disequilibrium. Raise one container by 2 cm: water flows from it to the others of the same good, the surfaces leave the line. This is the on-screen “imbalance”: at the wrong prices a buyer wants a different amount.

Incomes ×2. Press the button in the program: prices double, but the rim heights do not change — not a drop moves on the rig. Marginal utilities are the same, only money changes.

Harvest C +50%. Add another 220 ml to system C and reset the rims from the program's new table: the C rims drop (for I: 3.4 → 2.1 cm) — the good got cheaper, its marginal utility fell, and the water comes back to the line.

The rig is accurate to a few millimetres: the meniscus, imperfect walls and water in the tubes get in the way. That is enough to show the qualitative effects.